

PHYS 143 – Mechanics and Thermal Physics

Vectors

How do we describe, combine and calculate with quantities that have both magnitude and direction?

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- **Text:** Young and Freedman, *University Physics* (15th ed.). Copies in the Physics Department Library and Balme Library.
- **Sakai** (sakai.ug.edu.gh): slides, problem sheets, announcements, office hours. Log in this week.
- **Attendance** is required; ten absences without a UG physician's note bars you from the final exam.
- **Homework** must be your own work. Asking lecturers, TAs and classmates for help is encouraged.

Everything else

Venues, times, TA names and the assessment breakdown are on Sakai.

By the end of this week, you should be able to:

1. Represent, add and subtract vectors graphically.
2. Resolve vectors into components and reconstruct their magnitude and direction.
3. Add vectors using components and unit-vector notation.
4. Calculate and interpret dot and cross products.

Three contacts:

1 UNDERSTAND

Why do we need vectors?

2 CALCULATE

How do we calculate with vectors accurately?

3 EXTEND

How do vector products describe physical quantities?

Reading: Young and Freedman, Chapter 1, §1.7–1.10.

What to master, and what supports it

Core skills you should master

- distinguish scalars from vectors;
- add and subtract vectors graphically;
- resolve vectors into components;
- determine component signs;
- determine the correct quadrant;
- reconstruct magnitude and direction;
- add vectors using components;
- use unit-vector notation;
- calculate and interpret a dot product;
- calculate and interpret a cross product.

Supporting tools

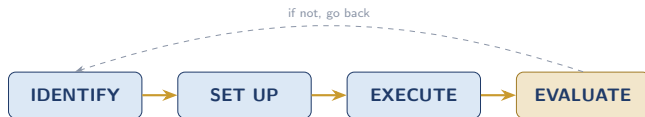
- detailed sine-rule and cosine-rule treatment;
- more elaborate vector-triangle cases;
- extended 3D material (supplement).

Supporting slides carry the label *supporting tool* in their titles. They stay in the deck for reference; the core skills are what the problems test first.

Contact 1: Understand

Why do we need vectors?

How we solve every problem this semester



IDENTIFY

What is being asked?
What quantities are known?
What physics is relevant?

SET UP

Draw a diagram.
Choose coordinates.
Predict the approximate size and direction.
Select the equations.

EXECUTE

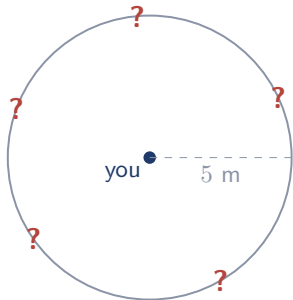
Perform the calculation.
Carry units.
Keep track of signs.

EVALUATE

Are the units correct?
Is the magnitude reasonable?
Does the direction agree with the sketch?
Does it make physical sense?

But what if... a bag is “5 m” from you. Is that enough to find it? We need to describe the quantities themselves, starting with **magnitude**.

Magnitude: how much?



Magnitude

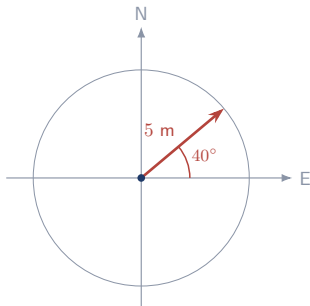
The **size** of a quantity: a number with units, such as 5 m, 70 kg, 15 m/s. It answers *how much?* and is never negative.

Predict first

A bag is 5 m from you. Which of the marked points is it?

But what if... the answer to *how much?* is not enough to locate the bag: every point on the circle is 5 m away. We also need **direction**.

Direction: which way?



Direction

The **orientation** of a quantity relative to a reference: a compass word, or an angle from a reference axis. It answers *which way?*

Rule: a complete answer to “where?”

magnitude *and* direction: 5 m at 40° north of east.

But what if... do all quantities need a direction? Mass is 70 kg whichever way you face. Some quantities need both, some only a size: **scalars and vectors**.

Scalar

A scalar quantity is **described completely by a numerical value and appropriate units**, where applicable.

e.g. mass, time, temperature, distance, speed, energy.

Notation: vector and magnitude

$\vec{A}$ = the vector

$A = |\vec{A}|$ = magnitude of the vector

In $\vec{A} \cdot \vec{B} = AB \cos \phi$, A and B are magnitudes.
 A_x is a number; $A_x \hat{i}$ is a vector.

Vector

A vector quantity **requires both magnitude and direction**.

e.g. displacement, velocity, acceleration, force.

Keep these distinct

The magnitude of a vector is a **scalar** and is **never negative**.

Concept check: scalar or vector?

Decide for each pair, then compare with your neighbour

- | | |
|---------------------------|-------------------------------|
| (a) 15 m/s | (b) 15 m/s north |
| (c) distance walked, 3 km | (d) displacement, 3 km east |
| (e) temperature, 30°C | (f) a force of 20 N downwards |
| (g) -5 J of energy | (h) mass, 70 kg |

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Answers

Scalars: (a) speed, (c) distance, (e), (g), (h). Vectors: (b) velocity, (d) displacement, (f) force.

A negative number is not automatically a vector: -5 J is still a scalar.

But what if... distance is a scalar. Walk 400 m round a track and stop where you began: does the distance tell you where you are? → **displacement**.

Distance and displacement

Distance

scalar
depends on the total
path travelled

Displacement

vector
runs directly from
initial position to final
position

Question

You walk once around a 400 m track and stop where you started. What is your displacement?

Distance and displacement

Distance

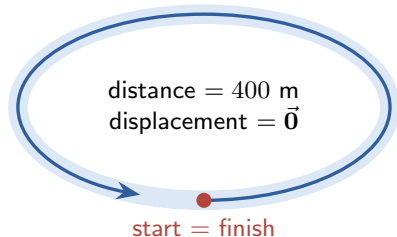
scalar
depends on the total
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Displacement

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Question

You walk once around a 400 m track and stop where you started. What is your displacement?



Zero, although the distance travelled is 400 m.

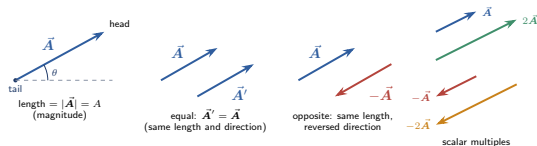
But what if... displacement is a vector. How do we name, draw and compare vectors? →
vocabulary.

Defining the vocabulary of vectors

Term	What it is	What it measures	Units	Representation
Magnitude	the size of a vector; a scalar, never negative	how much	those of the vector (m, m/s, N)	length of the arrow
Direction	the orientation of a vector in space	which way	degrees, or compass words	orientation of the arrow; angle θ
Displacement	straight line from the initial to the final position	how far, and which way, you are from the start	m	arrow from start to finish
Resultant	the single vector equal to a sum of vectors	the net effect of several vectors	those of the vectors added	arrow from first tail to last head

Interpret: magnitude answers *how much?*, direction answers *which way?*; a vector answers both. Component and unit vector are defined in Contact 2.

Representing a vector



Rule: equal and opposite vectors

Two vectors are **equal** if they have the same magnitude and the same direction, wherever they are drawn.

$-\vec{A}$ has the same magnitude as $\vec{A}$ and the opposite direction.

Rule: multiplying by a scalar

$c\vec{A}$ has magnitude $|c|A$.

$c > 0$: the direction is unchanged ($2\vec{A}$).

$c < 0$: the direction is reversed ($-\vec{A}$, $-2\vec{A}$).

But what if... two displacements follow one another, 1 km north then 2 km east. What single displacement results? → the **opening problem**.

An opening problem

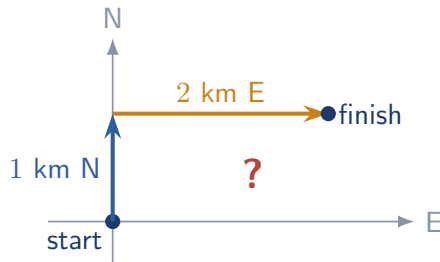
Opening problem

You walk **1 km north**, then **2 km east**.

Where are you now, relative to where you started?

Predict first

- Are you 3 km from the starting point?
- Is a single number a complete answer?
- In approximately what direction should the answer point?



We solve this once we can add vectors at 90° .

But what if... we cannot add 1 km and 2 km as numbers, because they point different ways. We need a rule that respects direction: **graphical addition**.

Adding two vectors: three equivalent ways

Predict first

If $\vec{A}$ and $\vec{B}$ are swapped, does the resultant change?



(a) $\vec{A}$ first, then $\vec{B}$



(b) $\vec{B}$ first, then $\vec{A}$

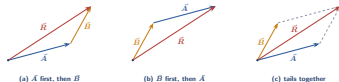


(c) tails together

Adding two vectors: three equivalent ways

Predict first

If $\vec{A}$ and $\vec{B}$ are swapped, does the resultant change?



Rule: head to tail, (a) and (b)

Put the tail of the second vector at the head of the first. $\vec{R}$ runs from the first tail to the last head.
 $\vec{A} + \vec{B} = \vec{B} + \vec{A}$ (commutative).

Rule: parallelogram, (c)

Draw both tails at one point. $\vec{R}$ is the *diagonal* from that point of the parallelogram built on $\vec{A}$ and $\vec{B}$.

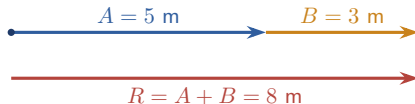
But what if... how big is $\vec{R}$? Start with the simplest cases, vectors along one line: **parallel** and **antiparallel**.

Predict first

Sketch 5 m east, then 3 m east. Predict the size and direction of the resultant *before* looking.

Predict first

Sketch 5 m east, then 3 m east. Predict the size and direction of the resultant *before* looking.



Rule: parallel vectors

Same direction: $|\vec{R}| = A + B$, in that common direction.

But what if... the second vector points the opposite way? Do the magnitudes still add?

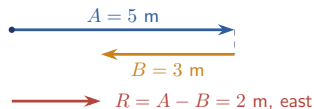
Predict first

Sketch 5 m east, then 3 m west. Predict the size and direction of the resultant *before* looking.

Antiparallel vectors

Predict first

Sketch 5 m east, then 3 m west. Predict the size and direction of the resultant *before* looking.



Rule: antiparallel vectors

Opposite directions: $|\vec{R}| = |A - B|$, in the direction of the *larger* vector.

These are the two extremes of $|\vec{A} + \vec{B}|$: largest at 0° between the vectors, smallest at 180° .

But what if... the vectors are not along one line at all? Is 3 m plus 4 m always 7 m?

Lecturer example: parallel and antiparallel

A hiker walks 4.5 km east, then a leg that is (i) 1.5 km east, (ii) 1.5 km west, (iii) 6.0 km west. Find $\vec{R}$.

Identify: net displacement $\vec{R}$; the legs lie on one line (parallel or antiparallel).

Set up: sketch each case on an east axis. Predict: (i) larger than 4.5, (ii) smaller than 4.5 but east, (iii) west.

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Execute (east = +):

(i) $R = 4.5 + 1.5 = 6.0$ km east

(ii) $R = 4.5 - 1.5 = 3.0$ km east

(iii) $R = 4.5 - 6.0 = -1.5$ km: 1.5 km west

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(iii) $R = 4.5 - 6.0 = -1.5$ km: 1.5 km west

Evaluate: units km; each result matches its prediction. In (iii) the larger vector is the westward one, so $\vec{R}$ points west. ✓

Your turn

Two forces on one block: 8 N east and 5 N west. Predict the size and direction of the net force, then calculate it.

Answer

3 N east.

Magnitudes do not simply add

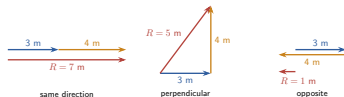
Predict first

Two vectors have magnitudes 3 m and 4 m. Must their resultant be 7 m?

Magnitudes do not simply add

Predict first

Two vectors have magnitudes 3 m and 4 m. Must their resultant be 7 m?



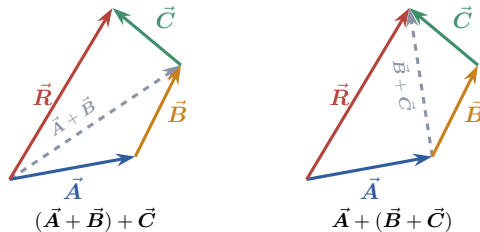
No. In general

$$|\vec{A} + \vec{B}| \neq |\vec{A}| + |\vec{B}|$$

The magnitudes add directly only when the vectors point in the same direction. For 3 m and 4 m, $1 \text{ m} \leq |\vec{A} + \vec{B}| \leq 7 \text{ m}$.

But what if... there are three or more vectors? Does the way we group them change $\vec{R}$?

Three or more vectors: associativity



Associative law

$(\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C})$. The vectors may be grouped in any way; the resultant is the same.

But what if... we want to undo an addition, or find the change between two vectors? → **subtraction**.

Subtracting vectors

Rule: subtraction

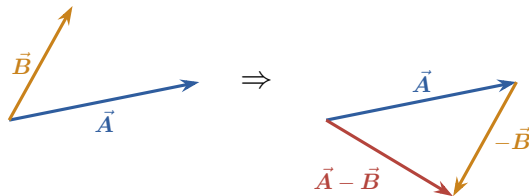
$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$

Reverse $\vec{B}$, then add head to tail.

Meaning: $\vec{A} - \vec{B}$ takes you from the head of $\vec{B}$ to the head of $\vec{A}$ (shared tail).

Connect: Lecture 2

Change in velocity $\Delta \vec{v} = \vec{v}_2 - \vec{v}_1$,
hence acceleration.



But what if... we need numbers, not drawings? A drawing is only as accurate as the ruler. Calculate, starting at 90° .

Concept check: before any numbers

Answer each with yes or no, and justify it with a sketch

1. Can a vector have a negative magnitude?
2. Can the displacement be zero while the distance travelled is not?
3. Can two vectors have the same magnitude but different directions? Are they equal?
4. Can $|\vec{A} + \vec{B}|$ be smaller than both $|\vec{A}|$ and $|\vec{B}|$?

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Answers

1. **No.** $|\vec{A}| \geq 0$; a minus sign in $-\vec{A}$ reverses the direction, it does not make the magnitude negative.
2. **Yes.** One lap of a track: displacement $\vec{0}$, distance 400 m.
3. **Yes, and they are not equal.** Equal vectors need the same magnitude *and* direction.
4. **Yes.** 5 m east plus 4 m west gives 1 m east.

TOOLKIT 1 – Directions and angles

WHEN TO USE: a direction is given in words, or must be reported as an angle.

PROCEDURE: $E = 0^\circ$, $N = 90^\circ$, $W = 180^\circ$, $S = 270^\circ$.

“ 30° north of east”: start east, turn north. “ 30° east of north”: start north, turn east.
Standard θ is measured counter-clockwise from $+x$. Always sketch before converting.

CHECK: the sketch and the angle θ must point the same way.

Predict first

Convert to θ : 30° east of north, 40° south of west, 20° west of south.

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CHECK: the sketch and the angle θ must point the same way.

Predict first

Convert to θ : 30° east of north, 40° south of west, 20° west of south.

$\theta = 60^\circ$, $\theta = 220^\circ$, $\theta = 250^\circ$.

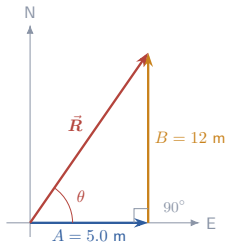
Predict first

You walk 5.0 m east, then 12 m north. Is the displacement 17 m? Predict its size and direction before calculating.

Adding two vectors at 90°

Predict first

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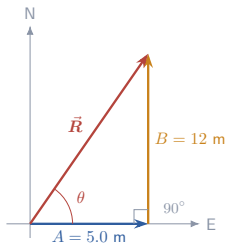
Head to tail, the two legs meet at a right angle, so $\vec{A}$, $\vec{B}$ and $\vec{R}$ form a **right triangle**, and $\vec{R}$ is its longest side (the hypotenuse).

But what if... we know R lies between 12 and 17 m, but exactly how long is it? Why would the hypotenuse follow a simple rule? → **Pythagoras**.

Adding two vectors at 90°

Predict first

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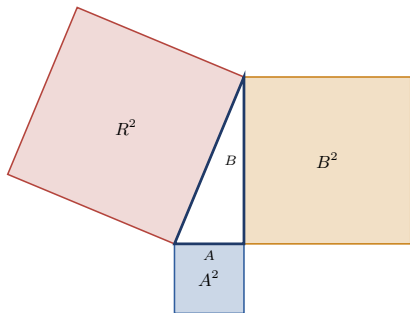


Head to tail, the two legs meet at a right angle, so $\vec{A}$, $\vec{B}$ and $\vec{R}$ form a **right triangle**, and $\vec{R}$ is its longest side (the hypotenuse).

Prediction: $12 < R < 17$ m (a straight line is shorter than the two legs), leaning toward north.

But what if... we know R lies between 12 and 17 m, but exactly how long is it? Why would the hypotenuse follow a simple rule? → **Pythagoras**.

Pythagoras: why it works



In a right triangle, the area of the square on the hypotenuse equals the sum of the areas of the squares on the other two sides.

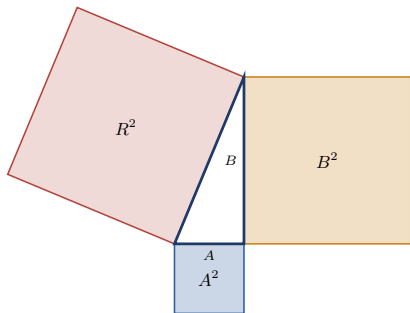
Rule: perpendicular vectors

$$R = \sqrt{A^2 + B^2}, \quad \theta = \tan^{-1} \frac{B}{A}$$

θ is the angle between $\vec{R}$ and $\vec{A}$.

But what if... this is the tool the opening problem needs. Does it also work when the legs are not at 90° ? First, solve the opening problem.

Pythagoras: why it works



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Rule: perpendicular vectors

$$R = \sqrt{A^2 + B^2}, \quad \theta = \tan^{-1} \frac{B}{A}$$

θ is the angle between $\vec{R}$ and $\vec{A}$.

$$R = \sqrt{5.0^2 + 12^2} = \sqrt{169} = 13 \text{ m},$$
$$\theta = \tan^{-1}(12/5.0) = 67^\circ \text{ north of east. Check:}$$
$$12 < 13 < 17, \text{ and } B > A \text{ so } \vec{R} \text{ leans north. } \checkmark$$

But what if... this is the tool the opening problem needs. Does it also work when the legs are not at 90° ? First, solve the opening problem.

TOOLKIT 2 – Vector triangles (right-angle case)

WHEN TO USE: two vectors are added head to tail and the legs meet at 90° .

PROCEDURE: **1.** Construct the head-to-tail triangle; $\vec{R}$ runs from the first tail to the last head. **2.** Check the angle between the legs. **3.** $R = \sqrt{A^2 + B^2}$, $\theta = \tan^{-1}(B/A)$.

CHECK: $|A - B| \leq R \leq A + B$; for 5.0 and 12 m, $7.0 \leq R \leq 17$ m, and 13 m passes.

WATCH OUT: the internal angle is not always the angle you were given. Next: when it is not 90° .

Predict first

For the opening problem (1 km north, 2 km east) what is the internal angle? Which tool applies?

Returning to the opening problem

Identify: asked for the displacement $\vec{R}$ (magnitude *and* direction); known: two perpendicular legs.

Set up: sketch head to tail, axes N and E. Predict: $2 < R < 3$ km, closer to east. Pythagoras and $\tan$.

Returning to the opening problem

Identify: asked for the displacement $\vec{R}$ (magnitude *and* direction); known: two perpendicular legs.

Set up: sketch head to tail, axes N and E. Predict: $2 < R < 3$ km, closer to east. Pythagoras and $\tan$.

Execute:

$$R = \sqrt{(1.00 \text{ km})^2 + (2.00 \text{ km})^2} = 2.24 \text{ km}$$

$$\tan \theta = \frac{2.00 \text{ km}}{1.00 \text{ km}} \Rightarrow \theta = 63.4^\circ \text{ east of north}$$

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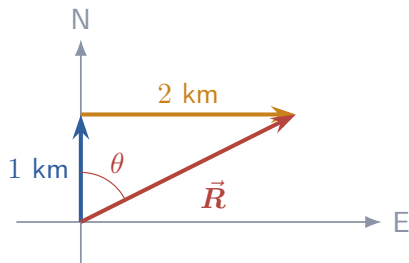
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$$\tan \theta = \frac{2.00 \text{ km}}{1.00 \text{ km}} \Rightarrow \theta = 63.4^\circ \text{ east of north}$$

Evaluate: units km; $2 < R < 3$ km as predicted; direction closer to east, matching the sketch.



Young, Example 1.5

But what if... change the second leg to 2.00 km at 60° north of east, after 3.00 km east? Pythagoras would give $\sqrt{3.00^2 + 2.00^2} = 3.61$ km. Is that right?

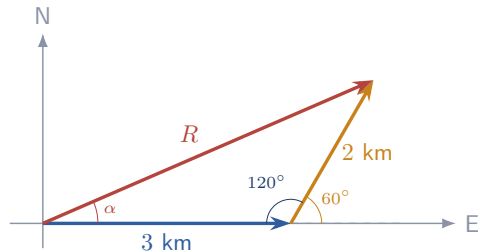
Limitation of Pythagoras: when the angle is not 90°

Your turn

A cyclist rides **3.00 km east**, then **2.00 km at 60° north of east**. Find the resultant displacement.

Predict first

Sketch it. Is Pythagoras still allowed? Is R closer to 5 km or to 3 km?



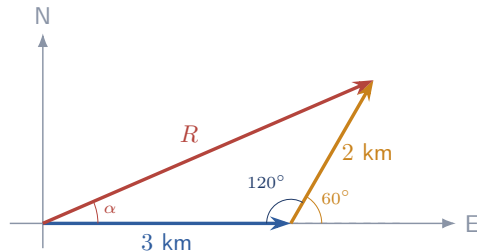
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Sketch it. Is Pythagoras still allowed? Is R closer to 5 km or to 3 km?



The triangle has no right angle: the angle *inside* the triangle between the two legs is $180^\circ - 60^\circ = 120^\circ$, not the 60° given. Pythagoras does not apply directly.

But what if... we need a tool for *any* triangle? → the **cosine and sine rules**.

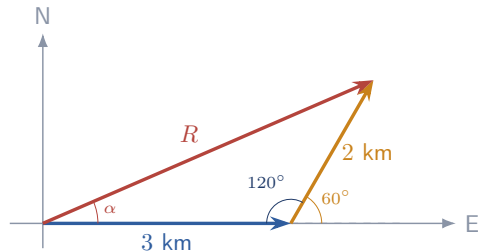
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Predict first

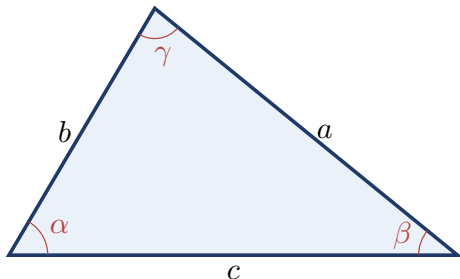
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The triangle has no right angle: the angle *inside* the triangle between the two legs is $180^\circ - 60^\circ = 120^\circ$, not the 60° given. Pythagoras does not apply directly. **Proof of failure:** Pythagoras gives 3.61 km, but the second leg alone carries you $2.00 \cos 60^\circ = 1.00$ km further east: at least 4.00 km from the start. So 3.61 km is impossible.

But what if... we need a tool for *any* triangle? → the **cosine and sine rules**.

A tool for vector triangles: the cosine and sine rules (supporting tool)



side a is opposite angle α , etc.

Cosine rule: find a side

$$a^2 = b^2 + c^2 - 2bc \cos \alpha$$

For two vectors, $R^2 = A^2 + B^2 - 2AB \cos \gamma$,
with γ the internal angle opposite $\vec{R}$.

Sine rule: find an angle

$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$$

These are mathematical tools, not a new vector idea; we use them without derivation. If $\alpha = 90^\circ$ the cosine rule becomes Pythagoras.

✂ TOOLKIT 2 – Vector triangles (completed)

WHEN TO USE: two vectors are added head to tail at any angle between the legs.

PROCEDURE: 1. Construct head to tail. 2. Find the internal angle γ opposite $\vec{R}$. 3. Right angle: Pythagoras. Otherwise the cosine rule for a side, $R^2 = A^2 + B^2 - 2AB \cos \gamma$, then the sine rule for an angle.

CHECK: $|A - B| \leq R \leq A + B$. For 3 m and 4 m, $1 \text{ m} \leq R \leq 7 \text{ m}$, so a calculated 9 m must be wrong.

WATCH OUT: γ is the internal angle of the triangle. It is not always the angle you were given.

Which angle is which? Triangle angle and vector-product angle

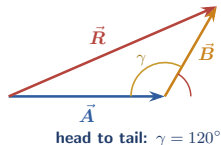
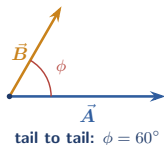
Predict first

In the cyclist problem the given angle is 60° . What is the internal angle γ opposite R ?

Which angle is which? Triangle angle and vector-product angle

Predict first

In the cyclist problem the given angle is 60° . What is the internal angle γ opposite R ?



Rule: two different angles

γ : internal angle of the head-to-tail triangle (120° , so the cosine rule uses $\cos 120^\circ$).

ϕ : tail-to-tail angle between the vectors, used later in dot and cross products (60° here).

The triangle angle γ and the vector-product angle ϕ are not necessarily the same angle.

Cyclist example: solution (supporting tool)

Set up: sketch; the 120° angle is opposite R . Predict $3 < R < 5$ km.

Execute, cosine rule:

$$R^2 = 3.00^2 + 2.00^2 - 2(3.00)(2.00) \cos 120^\circ = 19.0$$

$$R = 4.36 \text{ km}$$

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sine rule for the direction α at the start:

$$\frac{\sin \alpha}{2.00} = \frac{\sin 120^\circ}{4.36} \Rightarrow \alpha = 23.4^\circ$$

$\vec{R} = 4.36 \text{ km}$ at 23.4° north of east.

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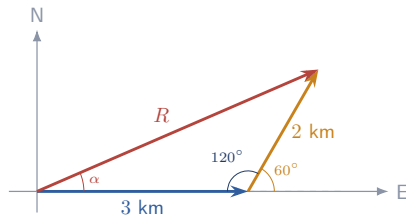
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$$R = 4.36 \text{ km}$$

sine rule for the direction α at the start:

$$\frac{\sin \alpha}{2.00} = \frac{\sin 120^\circ}{4.36} \Rightarrow \alpha = 23.4^\circ$$

$\vec{R} = 4.36$ km at 23.4° north of east. **Evaluate:** units km; $3 < R < 5$ as predicted; $0^\circ < \alpha < 60^\circ$, matching the sketch. ✓



But what if... the rider makes four legs? Each leg needs another triangle, with its own cosine and sine rule. Is there a systematic method? → **components**.

Question

A delivery rider makes four legs: 2.1 km at 20° N of E, 3.4 km at 35° W of N, 1.8 km S, 2.6 km at 10° S of W. Where does the rider end up?

- Graphical methods give physical understanding ...
- ... but scale drawings have limited precision,
- and repeated triangle calculations become cumbersome.
- **Components** provide a systematic method for any number of vectors.

Next time

Components turn a geometrical vector problem into an algebraic one.

Common mistakes and the correct habit

Mistake

Giving the magnitude a sign: $|\vec{A}| < 0$.

Adding magnitudes: $|\vec{A} + \vec{B}| = |\vec{A}| + |\vec{B}|$.

Calling any negative number a vector (-5 J).

Using Pythagoras when the angle is not 90° .

Reading “ 30° north of east” from north.

Confusing distance with displacement.

Instead

The magnitude is never negative; the minus sign belongs to the direction.

Add head to tail; then $|A - B| \leq |\vec{A} + \vec{B}| \leq A + B$.

Ask whether a direction is needed: energy is a scalar.

Find the internal angle, then the cosine and sine rules.

Start from the first named direction and turn toward the second.

Displacement is final minus initial; distance is the path length.

Contact 2: Calculate

How do we calculate with vectors accurately?

Graphs for geometric understanding, components for calculation

Graphical methods

- Show what is happening
- Good for sketches and checks
- Limited accuracy; awkward for many vectors

Components

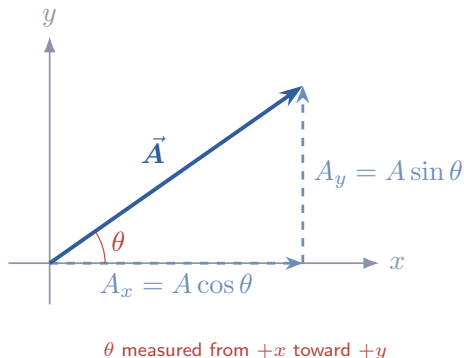
- Turn vectors into ordinary numbers
- Any number of vectors, any accuracy
- Work the same way in 3D

Physical meaning

Graphs give geometric understanding; components provide a systematic way to calculate accurately. Always sketch first, predict, calculate, then check against the sketch.

But what if... we describe a vector by how much of it lies along each axis? What is that “how much”, and how do we find it? → **components**.

What is a component?



Define. A **component** is the part of a vector along one chosen axis. A_x tells us **how much of $\vec{A}$ lies along the x -direction**; A_y , **how much along the y -direction**. Units: those of the vector. Drawn as a projection on the axis.

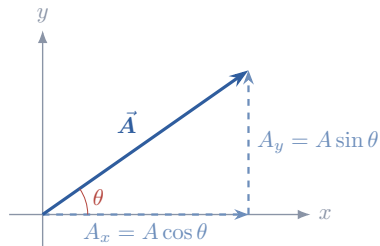
Scalar numbers, not vectors

A component such as A_x is a number (positive, negative or zero), as are A_y , A_z . **The corresponding component vector is $A_x \hat{i}$, so**

$$\vec{A} = A_x \hat{i} + A_y \hat{j}.$$

But what if... we know only the magnitude A and the angle θ ? How do we get A_x and A_y ? $\rightarrow$ **resolving.**

Components of a vector



θ measured from $+x$ toward $+y$

Rule: components (2D)

$$A_x = A \cos \theta, \quad A_y = A \sin \theta$$

These apply when θ is measured from the $+x$ -axis toward the $+y$ -axis.

Number or vector?

A component such as A_x is a number. The corresponding component vector is $A_x \hat{i}$.

EQUATION – components

Meaning. A_x , A_y give how much of $\vec{A}$ lies along each axis.

Variables. A magnitude; θ angle from $+x$.

Units. A_x , A_y carry the units of $\vec{A}$; θ is in degrees.

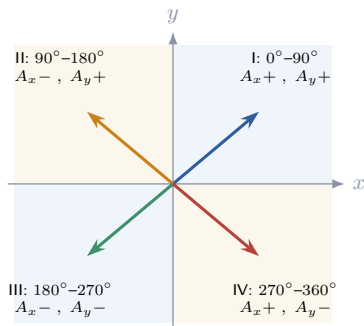
Interpretation. the sign says right or left, up or down; $|A_x| \leq A$.

Predict first

For $\theta = 150^\circ$, is A_x positive or negative? Is A_y ?

But what if... the vector points left or down, as at 150° ? Then $\cos \theta$ or $\sin \theta$ is negative. $\rightarrow$ quadrants.

Signs and quadrants: predict first



Before using equations, ask: does the vector point **right or left**? Does it point **up or down**?

Sign rule

quadrant	I	II	III	IV
A_x	> 0	< 0	< 0	> 0
A_y	> 0	> 0	< 0	< 0

Permanent rule

Sketch → **predict signs** → **calculate**
The calculator never replaces the sketch.

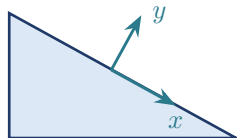
But what if... the angle is given from another axis, such as 37° from $+y$ or 45° below $+x$? Then $A \cos \theta$ and $A \sin \theta$ swap roles. → **resolving carefully**.

✂ TOOLKIT 3 – Components

WHEN TO USE: a vector is given by a magnitude and an angle, and you need its parts along chosen axes.

PROCEDURE: Choose axes (perpendicular, not necessarily horizontal and vertical). **Sketch and predict signs:** right or left gives the sign of A_x ; up or down gives the sign of A_y . **Resolve:** $A_x = A \cos \theta$, $A_y = A \sin \theta$ for θ from $+x$; otherwise *adjacent* $\rightarrow$ *cosine*, *opposite* $\rightarrow$ *sine*, with signs from the sketch.

CHECK: the signs match the sketch and $\sqrt{A_x^2 + A_y^2} = A$. A_x is a number; $A_x \hat{i}$ is the component vector.



axes tilted with a slope

Decide, then compare with your neighbour

1. Can a component be negative? Can it be zero for a non-zero vector?
2. Can $|A_x|$ be larger than the magnitude A ?
3. If you rotate the axes, does the vector change? Do its components change?

Decide, then compare with your neighbour

1. Can a component be negative? Can it be zero for a non-zero vector?
2. Can $|A_x|$ be larger than the magnitude A ?
3. If you rotate the axes, does the vector change? Do its components change?

Answers

1. **Yes to both.** The sign gives the direction along the axis; a vector along y has $A_x = 0$.
2. **No.** $A_x = A \cos \theta$ and $|\cos \theta| \leq 1$.
3. **The vector does not change; its components do.** Components describe the vector *relative to chosen axes*, which is why we choose axes to make a problem simple.

Question

A force of 12.0 N acts at $\theta = 150^\circ$ (counter-clockwise from $+x$). Find F_x and F_y .

Sketch $\rightarrow$ predict signs $\rightarrow$ calculate

Right or left? Up or down? So what signs do you expect?

Question

A force of 12.0 N acts at $\theta = 150^\circ$ (counter-clockwise from $+x$). Find F_x and F_y .

Sketch $\rightarrow$ **predict signs** $\rightarrow$ **calculate**

Right or left? Up or down? So what signs do you expect?

Quadrant II, so $F_x < 0$, $F_y > 0$.

$$F_x = 12.0 \cos 150^\circ = -10.4 \text{ N}$$

$$F_y = 12.0 \sin 150^\circ = +6.00 \text{ N}$$

Evaluate: $\sqrt{10.4^2 + 6.00^2} = 12.0 \checkmark$

Your turn: angles measured from other axes

Your turn

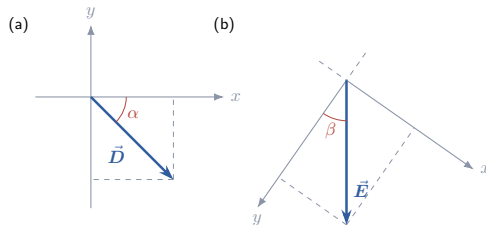
Find the x - and y -components of $\vec{D}$ and $\vec{E}$.

(a) $|\vec{D}| = 3.00 \text{ m}$, $\alpha = 45^\circ$ below $+x$.

(b) $|\vec{E}| = 4.50 \text{ m}$, $\beta = 37.0^\circ$ from the $+y$ axis, with tilted axes.

Sketch $\rightarrow$ **predict signs** $\rightarrow$ **calculate**.

Young, Example 1.6.



Your turn: solution

(a) Measured counter-clockwise from $+x$,
 $\theta = -45^\circ$:

$$D_x = 3.00 \cos(-45^\circ) = +2.12 \text{ m}$$

$$D_y = 3.00 \sin(-45^\circ) = -2.12 \text{ m}$$

Your turn: solution

(a) Measured counter-clockwise from $+x$,
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(b) β is measured from $+y$, so E_y is *adjacent*
and E_x is *opposite*:

$$E_x = 4.50 \sin 37.0^\circ = +2.71 \text{ m}$$

$$E_y = 4.50 \cos 37.0^\circ = +3.59 \text{ m}$$

Both positive, since $\vec{E}$ lies between the $+x$ and $+y$ axes of the tilted system.

Your turn: solution

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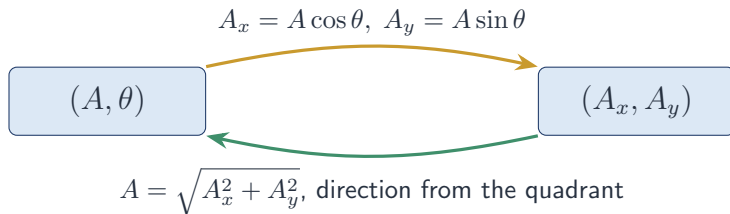
Both positive, since $\vec{E}$ lies between the $+x$ and $+y$ axes of the tilted system.

Physical meaning

The formulas $A \cos \theta$, $A \sin \theta$ only hold for θ measured from $+x$. Otherwise think “adjacent or opposite?”

But what if... we go backwards: given A_x and A_y , find A and θ ? A calculator gives the same $\tan^{-1}$ for two vectors that point opposite ways. $\rightarrow$ **reconstructing**.

Two equivalent descriptions of one vector



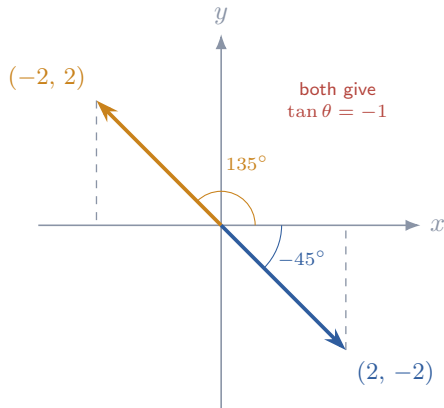
Key idea

Magnitude/direction and components are two equivalent ways of describing the same vector. Going from components back to magnitude and direction is Toolkit 4.

Direction from components: a common error

Predict first

$\vec{P} = (-2, 2)$ and $\vec{Q} = (2, -2)$. Both have $A_y/A_x = -1$, and a calculator gives $\tan^{-1}(-1) = -45^\circ$ for both. Same direction?



Finding direction from components

🔧 TOOLKIT 4 – Reconstructing a vector

WHEN TO USE: the components are known and you need the magnitude and direction.

PROCEDURE: $A = \sqrt{A_x^2 + A_y^2}$. **1.** Quadrant from the signs of A_x , A_y . **2.** Reference angle $\theta_{\text{ref}} = \tan^{-1} |A_y/A_x|$. **3.** Place it: I: θ_{ref} ; II: $180^\circ - \theta_{\text{ref}}$; III: $180^\circ + \theta_{\text{ref}}$; IV: $360^\circ - \theta_{\text{ref}}$.

CHECK: the direction agrees with the sketch.

WATCH OUT: vectors with the same A_y/A_x can point in different directions. **Quadrant first, calculation second.**

$\vec{P} = (-2, 2)$: left and up, quadrant II.
 $\theta_{\text{ref}} = \tan^{-1}(2/2) = 45^\circ$

$$\theta_P = 180^\circ - 45^\circ = 135^\circ$$

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$\vec{Q} = (2, -2)$: right and down, quadrant IV.
 $\theta_{\text{ref}} = 45^\circ$

$$\theta_Q = 360^\circ - 45^\circ = 315^\circ (= -45^\circ)$$

Finding direction from components

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Your turn: $A_x = -3.0$ m, $A_y = -4.0$ m. Find A and θ .

But what if... there are several vectors? How do their components combine? → **adding vectors.**

Finding direction from components

🔧 TOOLKIT 4 – Reconstructing a vector

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 $\theta_{\text{ref}} = 45^\circ$

$$\theta_Q = 360^\circ - 45^\circ = 315^\circ (= -45^\circ)$$

Your turn: $A_x = -3.0$ m, $A_y = -4.0$ m. Find A and θ . $A = 5.0$ m; quadrant III;
 $\theta_{\text{ref}} = \tan^{-1}(4/3) = 53.1^\circ$;
 $\theta = 180^\circ + 53.1^\circ = 233.1^\circ$ ✓

But what if... there are several vectors? How do their components combine? → **adding vectors.**

Question

If $\vec{R} = \vec{A} + \vec{B} + \vec{C} + \dots$, how are the components of $\vec{R}$ related to theirs?

Adding vectors with components

Question

If $\vec{R} = \vec{A} + \vec{B} + \vec{C} + \dots$, how are the components of $\vec{R}$ related to theirs?

Rule: adding vectors by components

$$R_x = \sum A_x, \quad R_y = \sum A_y \quad (\text{in 3D also } R_z = \sum A_z)$$

✂ TOOLKIT 5 – Adding vectors by components

WHEN TO USE: two or more vectors must be added, whatever their directions.

PROCEDURE: **RESOLVE** each vector (a table helps), **COMBINE** like components, **RECONSTRUCT** the magnitude and direction of $\vec{R}$ (Toolkit 4). **Sketch** → **predict signs** → **calculate**.

CHECK: $\vec{R}$ lies in the quadrant predicted by the sketch, and R is no larger than the sum of the magnitudes.

But what if... the vectors are given as magnitudes and compass directions? → **multiple vectors**.

Multiple vectors: three displacements (Young Ex. 1.7)

Question

A contestant in a radio race makes three displacements:

$\vec{A}$: 72.4 m, 32.0° **east of north**

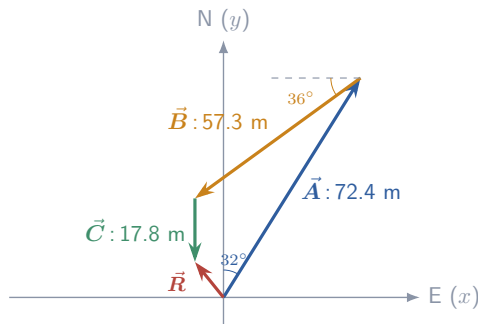
$\vec{B}$: 57.3 m, 36.0° **south of west**

$\vec{C}$: 17.8 m **south**

Find the resultant $\vec{R}$.

Predict first

From the sketch: roughly how far, and in which quadrant?



Identify: target $\vec{R}$; known three displacements. **Set up:** x east, y north; convert each direction to θ from $+x$: $\theta_A = 90^\circ - 32.0^\circ = 58.0^\circ$, $\theta_B = 180^\circ + 36.0^\circ = 216.0^\circ$, $\theta_C = 270^\circ$.

Identify: target $\vec{R}$; known three displacements. **Set up:** x east, y north; convert each direction to θ from $+x$: $\theta_A = 90^\circ - 32.0^\circ = 58.0^\circ$, $\theta_B = 180^\circ + 36.0^\circ = 216.0^\circ$, $\theta_C = 270^\circ$.

	magnitude	θ	x -component	y -component
$\vec{A}$	72.4 m	58.0°	38.37 m	61.40 m
$\vec{B}$	57.3 m	216.0°	-46.36 m	-33.68 m
$\vec{C}$	17.8 m	270.0°	0.00 m	-17.80 m
$\vec{R}$			$R_x = -7.99 \text{ m}$	$R_y = 9.92 \text{ m}$

Identify: target $\vec{R}$; known three displacements. **Set up:** x east, y north; convert each direction to θ from $+x$: $\theta_A = 90^\circ - 32.0^\circ = 58.0^\circ$, $\theta_B = 180^\circ + 36.0^\circ = 216.0^\circ$, $\theta_C = 270^\circ$.

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$\vec{C}$	17.8 m	270.0°	0.00 m	-17.80 m
$\vec{R}$			$R_x = -7.99 \text{ m}$	$R_y = 9.92 \text{ m}$

Execute (resolve, combine, reconstruct): $R = \sqrt{(-7.99)^2 + 9.92^2} = 12.7 \text{ m}$; left and up, quadrant II; $\theta_{\text{ref}} = \tan^{-1}\left(\frac{9.92}{7.99}\right) = 51.1^\circ$, so $\theta = 180^\circ - 51.1^\circ = 129^\circ$, i.e. 12.7 m at 39° west of north. **Evaluate:** units m; quadrant II, matching the sketch. ✓

Stating the direction in words

A calculation may give $\theta = 129^\circ$, but the physical answer is often clearer as a compass direction.

Measure from the nearest cardinal axis:

$$\text{north} = 90^\circ, \quad 129^\circ - 90^\circ = 39^\circ$$

so $\theta = 129^\circ$ is 39° **west of north**. Check by drawing it.

Your turn

Express $\theta = 233.1^\circ$ and $\theta = 96.2^\circ$ as compass directions.

Stating the direction in words

A calculation may give $\theta = 129^\circ$, but the physical answer is often clearer as a compass direction.

Measure from the nearest cardinal axis:

$$\text{north} = 90^\circ, \quad 129^\circ - 90^\circ = 39^\circ$$

so $\theta = 129^\circ$ is 39° **west of north**. Check by drawing it.

Your turn

Express $\theta = 233.1^\circ$ and $\theta = 96.2^\circ$ as compass directions.

$233.1^\circ - 180^\circ = 53.1^\circ$ south of west;
 $96.2^\circ - 90^\circ = 6.2^\circ$ west of north.

Independent problem

A hiker walks 4.00 km at 30.0° north of east, then 6.00 km at 40.0° west of north, then 3.00 km south. Find the magnitude and direction of the hiker's resultant displacement.

Predict first

Sketch first. Which quadrant will $\vec{R}$ be in?

Independent problem

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Predict first

Sketch first. Which quadrant will $\vec{R}$ be in?

$$\theta: 30.0^\circ, 130.0^\circ, 270^\circ$$

$$R_x = 3.46 - 3.86 + 0 = -0.39 \text{ km}$$

$$R_y = 2.00 + 4.60 - 3.00 = 3.60 \text{ km}$$

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$$R_x = 3.46 - 3.86 + 0 = -0.39 \text{ km}$$

$$R_y = 2.00 + 4.60 - 3.00 = 3.60 \text{ km}$$

$$R = 3.62 \text{ km}$$

left and up: quadrant II

$$\theta_{\text{ref}} = \tan^{-1}(3.60/0.39) = 83.8^\circ$$

$\theta = 180^\circ - 83.8^\circ = 96.2^\circ$, i.e. 6.2° west of north.

But what if... writing components in words is long? $\rightarrow$ **unit vectors**.

Unit vectors

Unit vectors $\hat{i}, \hat{j}, \hat{k}$

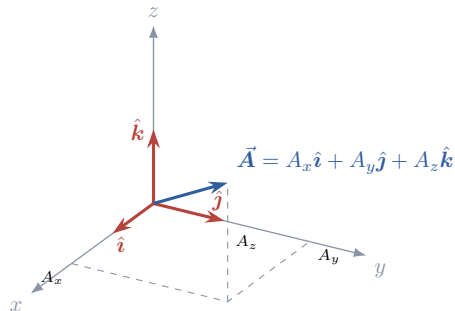
Direction only: magnitude 1, no units, along $+x, +y, +z$.

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

A compact way to write components *together with their directions*.

Number or vector?

A_x, A_y, A_z are **numbers**; $A_x \hat{i}, A_y \hat{j}, A_z \hat{k}$ are **component vectors**.



Rule: adding in unit-vector form

Collect like terms:

$$\begin{aligned}\vec{A} + \vec{B} &= (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} \\ &\quad + (A_z + B_z)\hat{k}\end{aligned}$$

Predict first

$\vec{A} = (2\hat{i} + 3\hat{j})$ m and $\vec{B} = (\hat{i} - 5\hat{j})$ m. Is $\vec{A} + \vec{B}$ up or down, left or right of the origin?

Adding vectors in unit-vector form

Rule: adding in unit-vector form

Collect like terms:

$$\begin{aligned}\vec{A} + \vec{B} &= (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} \\ &\quad + (A_z + B_z)\hat{k}\end{aligned}$$

Predict first

$\vec{A} = (2\hat{i} + 3\hat{j})$ m and $\vec{B} = (\hat{i} - 5\hat{j})$ m. Is $\vec{A} + \vec{B}$ up or down, left or right of the origin?

$\vec{A} + \vec{B} = (3\hat{i} - 2\hat{j})$ m: right and down, quadrant IV. $|\vec{A} + \vec{B}| = \sqrt{3^2 + (-2)^2} = 3.6$ m. ✓

Question

$\vec{D} = (6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k})$ m and $\vec{E} = (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k})$ m. Find $\vec{F} = 2\vec{D} - \vec{E}$.

Question

$\vec{D} = (6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k})$ m and $\vec{E} = (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k})$ m. Find $\vec{F} = 2\vec{D} - \vec{E}$.

$$\vec{F} = 2(6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k}) - (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k}) = (8.00\hat{i} + 11.00\hat{j} - 10.00\hat{k}) \text{ m}$$

$$F = \sqrt{(8.00 \text{ m})^2 + (11.00 \text{ m})^2 + (-10.00 \text{ m})^2} = 16.9 \text{ m}$$

Question

$\vec{D} = (6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k})$ m and $\vec{E} = (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k})$ m. Find $\vec{F} = 2\vec{D} - \vec{E}$.

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$$F = \sqrt{(8.00 \text{ m})^2 + (11.00 \text{ m})^2 + (-10.00 \text{ m})^2} = 16.9 \text{ m}$$

Your turn

$\vec{A} = (4\hat{i} - 5\hat{j} + 7\hat{k})$ m, $\vec{B} = (\hat{i} - 3\hat{j} + 5\hat{k})$ m, $\vec{C} = (\hat{i} + \hat{j} - 2\hat{k})$ m. Find $\vec{A} + \vec{B} + \vec{C}$ and its magnitude.

But what if... all examples so far are planar? Real forces are three-dimensional. $\rightarrow$ **3D vectors.**

Question

$\vec{D} = (6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k})$ m and $\vec{E} = (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k})$ m. Find $\vec{F} = 2\vec{D} - \vec{E}$.

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$$F = \sqrt{(8.00 \text{ m})^2 + (11.00 \text{ m})^2 + (-10.00 \text{ m})^2} = 16.9 \text{ m}$$

Your turn

$\vec{A} = (4\hat{i} - 5\hat{j} + 7\hat{k})$ m, $\vec{B} = (\hat{i} - 3\hat{j} + 5\hat{k})$ m, $\vec{C} = (\hat{i} + \hat{j} - 2\hat{k})$ m. Find $\vec{A} + \vec{B} + \vec{C}$ and its magnitude.
 $(6\hat{i} - 7\hat{j} + 10\hat{k})$ m, $|\cdot| = \sqrt{185} = 13.6$ m

But what if... all examples so far are planar? Real forces are three-dimensional. $\rightarrow$ **3D vectors.**

The 3D Cartesian system

Rule: right-handed axes

Curl the fingers of the right hand from $+x$ to $+y$; the thumb points along $+z$.

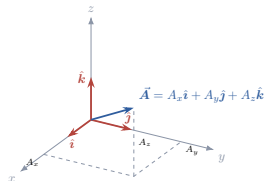
Rule: components and magnitude in 3D

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

Physical meaning

(Extended material, see the supplement.) The mutually perpendicular unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$ provide three independent directions, so a vector is uniquely specified by its x -, y - and z -components.



But what if... we can only add and subtract? How do we multiply vectors? → **Contact 3.**

Common mistakes and the correct habit

Mistake

Trusting $\tan^{-1}$ for the direction.

Using $A \cos \theta$, $A \sin \theta$ when θ is not from $+x$.

Adding magnitudes instead of components.

Calculating before predicting the signs.

Mixing scalar components with component vectors.

Calculator in the wrong angle mode.

Instead

Quadrant first from the signs, then the reference angle.

Ask adjacent or opposite, and take the signs from the sketch.

Resolve, combine, reconstruct.

Sketch, predict the signs, then calculate.

A_x is a number; $A_x \hat{i}$ is the vector.

Check degrees against radians before every trigonometric step.

Lecture 2: motion

Projectile motion: the x - and y -motions are treated separately, using

$$v_x = v \cos \theta \text{ and } v_y = v \sin \theta.$$

Lecture 3: forces

Newton's second law is applied component by component, $\sum F_x = ma_x$ and $\sum F_y = ma_y$, with axes along and perpendicular to a slope.

Beyond mechanics

Electric and magnetic fields and every other vector quantity are added by components in exactly the same way.

Physical meaning

Resolve, combine, reconstruct is a skill you will use every week of this course.

Contact 3: Extend

How do vector products describe physical quantities?

From combining vectors to constructing quantities

So far, we have used vectors to represent and combine physical quantities. We now ask a new question: **what useful physical quantities can we construct by combining two vectors?**

From combining vectors to constructing quantities

So far, we have used vectors to represent and combine physical quantities. We now ask a new question: **what useful physical quantities can we construct by combining two vectors?**

Pulling a crate

pulling a crate $\rightarrow$ **dot product** $\rightarrow$
work

Turning a door

turning a door $\rightarrow$ **cross product** $\rightarrow$
torque

Why multiply vectors? Two physical questions

Motivate: pulling a crate

A rope pulls a crate along the floor at an angle. **How much of the pull acts along the motion?**

We need one number that measures how much one vector points along another.

Dot product

vector $\cdot$ vector $\rightarrow$ **scalar**
leads to *work*

Motivate: turning a door

A push on a door turns it about its hinge. The effect depends on where and which way you push, and has an **axis**.

We need a vector, perpendicular to both position and force, that records the turning.

Cross product

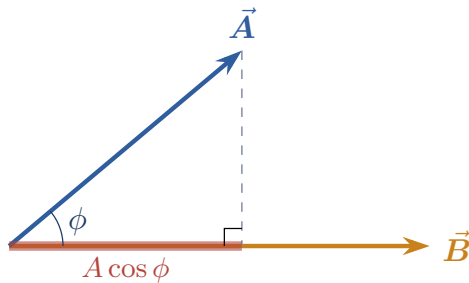
vector $\times$ vector $\rightarrow$ **vector**
leads to *torque*

We first understand what each product means geometrically, then learn how to calculate it.

The dot (scalar) product

Question

How much of one vector lies along the direction of another?



The part of $\vec{A}$ that lies along $\vec{B}$ is the **projection**

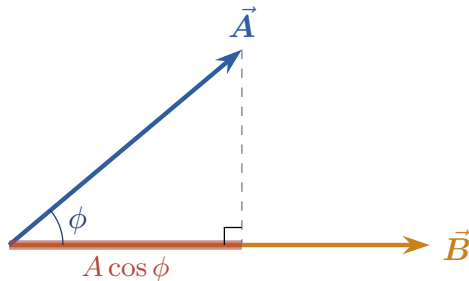
$$A \cos \phi$$

where **for dot and cross products**, ϕ is the **angle between the vectors when placed tail-to-tail**, $0^\circ \leq \phi \leq 180^\circ$.

The dot (scalar) product

Question

How much of one vector lies along the direction of another?



The part of $\vec{A}$ that lies along $\vec{B}$ is the **projection**

$$A \cos \phi$$

where **for dot and cross products**, ϕ is the **angle between the vectors when placed tail-to-tail**, $0^\circ \leq \phi \leq 180^\circ$.

Definition: dot product

$$\vec{A} \cdot \vec{B} = AB \cos \phi = (A \cos \phi) B$$

vector $\cdot$ vector $\rightarrow$ **scalar**. Units multiply:
 $\text{m} \cdot \text{m} = \text{m}^2$.

The angle ϕ : tail to tail

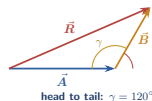
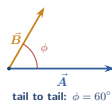
Predict first

The cyclist's legs are 3.00 km east and 2.00 km at 60° north of east. Is the angle ϕ for a dot or cross product 60° or 120° ?

The angle ϕ : tail to tail

Predict first

The cyclist's legs are 3.00 km east and 2.00 km at 60° north of east. Is the angle ϕ for a dot or cross product 60° or 120° ?



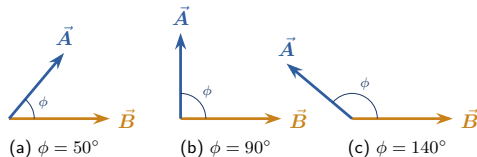
Rule: the angle ϕ

For dot and cross products, ϕ is the angle between the vectors when placed tail-to-tail. Here $\phi = 60^\circ$.

Common error

Do not use the internal angle of a head-to-tail vector triangle as ϕ . That angle is $\gamma = 120^\circ$.

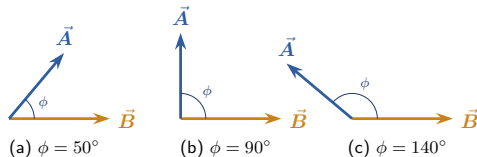
Predict the sign of $\vec{A} \cdot \vec{B}$



Predict first

For each pair: is $\vec{A} \cdot \vec{B}$ positive, zero, or negative?

Predict the sign of $\vec{A} \cdot \vec{B}$



Predict first

For each pair: is $\vec{A} \cdot \vec{B}$ positive, zero, or negative?

Rule: sign of the dot product

$$\phi < 90^\circ \Rightarrow \vec{A} \cdot \vec{B} > 0; \quad \phi = 90^\circ \Rightarrow \vec{A} \cdot \vec{B} = 0; \quad \phi > 90^\circ \Rightarrow \vec{A} \cdot \vec{B} < 0$$

Question

What are $\hat{i} \cdot \hat{i}$ and $\hat{i} \cdot \hat{j}$?

Question

What are $\hat{i} \cdot \hat{i}$ and $\hat{i} \cdot \hat{j}$?

Rule: unit-vector dot products

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1, \quad \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$$

Question

What are $\hat{i} \cdot \hat{i}$ and $\hat{i} \cdot \hat{j}$?

Rule: unit-vector dot products

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1, \quad \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$$

Expanding $(A_x\hat{i} + A_y\hat{j} + A_z\hat{k}) \cdot (B_x\hat{i} + B_y\hat{j} + B_z\hat{k})$, only the like terms survive:

Rule: dot product from components

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Connect: work, Lecture 4

$$W = \vec{F} \cdot \vec{s}$$

A constant force $\vec{F}$ acting through a displacement $\vec{s}$. Work is studied fully in Lecture 4.

EQUATION – dot product

Meaning. $\vec{A} \cdot \vec{B}$ measures how much $\vec{A}$ points along $\vec{B}$, scaled by the size of $\vec{B}$.

Variables. A, B are the magnitudes; ϕ is the angle between the vectors tail to tail, $0^\circ \leq \phi \leq 180^\circ$; $A_x, \dots, B_z$ are components.

Units. the product of the units: $\text{m} \times \text{m} = \text{m}^2$; for force and displacement, $\text{Nm} = \text{J}$.

Interpretation. positive: the vectors broadly agree in direction; zero: perpendicular; negative: they broadly oppose. The result is a **scalar**.

Example. the next slide, then the pair $\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}$, $\vec{B} = -5\hat{i} + \hat{j} - \hat{k}$.

Two forms, one scalar

$$\vec{A} \cdot \vec{B} = AB \cos \phi$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Predict first

Which form is easier when you have a diagram and two angles? Which when you have components?

Question

$|\vec{A}| = 7.50 \text{ m}$ at 53.0° , $|\vec{B}| = 5.00 \text{ m}$ at 130.0° . Find $\vec{A} \cdot \vec{B}$.

Predict first

Is ϕ less than or more than 90° ? So what sign?

Lecturer example: dot product from a figure

Question

$|\vec{A}| = 7.50 \text{ m}$ at 53.0° , $|\vec{B}| = 5.00 \text{ m}$ at 130.0° . Find $\vec{A} \cdot \vec{B}$.

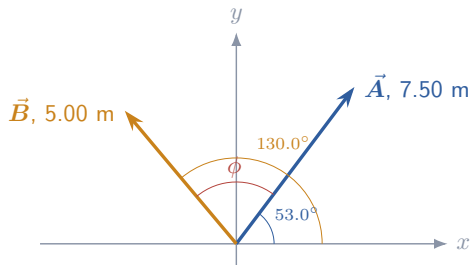
Predict first

Is ϕ less than or more than 90° ? So what sign?

$$\phi = 130.0^\circ - 53.0^\circ = 77.0^\circ$$

$$\vec{A} \cdot \vec{B} = (7.50 \text{ m})(5.00 \text{ m}) \cos 77.0^\circ = 8.44 \text{ m}^2$$

Evaluate: positive since $\phi < 90^\circ$; units are m^2 , not m.



after Young, Example 1.9

TOOLKIT 6 – Dot product

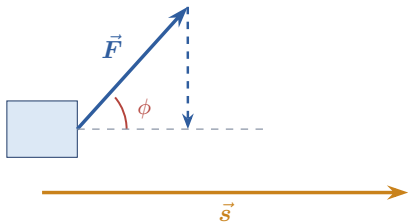
WHEN TO USE: you need a scalar: how much of one vector lies along another, the angle between two vectors, or work.

PROCEDURE: Two forms, one quantity: $\vec{A} \cdot \vec{B} = AB \cos \phi = A_x B_x + A_y B_y + A_z B_z$. Predict the sign: acute angle $\rightarrow$ positive; perpendicular $\rightarrow$ zero; obtuse $\rightarrow$ negative. Angle:

$\cos \phi = \frac{\vec{A} \cdot \vec{B}}{AB}$, with A, B the magnitudes. Preview: work, $W = \vec{F} \cdot \vec{s}$.

CHECK: the sign agrees with the geometry (positive $\rightarrow \phi < 90^\circ$, negative $\rightarrow \phi > 90^\circ$), and ϕ is the tail-to-tail angle, not γ .

Work: the dot product in action



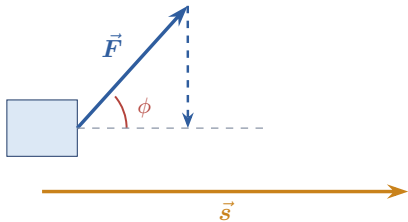
Lecturer example

A rope pulls a crate with $F = 30 \text{ N}$ at $\phi = 40^\circ$ above the horizontal while the crate moves $s = 5.0 \text{ m}$. Find the work done by the rope.

Predict first

Positive or negative? Larger or smaller than $Fs = 150 \text{ J}$?

Work: the dot product in action



Lecturer example

A rope pulls a crate with $F = 30 \text{ N}$ at $\phi = 40^\circ$ above the horizontal while the crate moves $s = 5.0 \text{ m}$. Find the work done by the rope.

Predict first

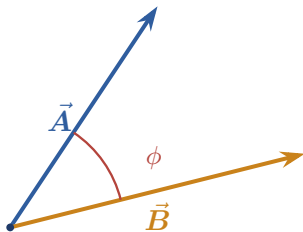
Positive or negative? Larger or smaller than $Fs = 150 \text{ J}$?

Identify: $W = \vec{F} \cdot \vec{s}$. **Set up:** ϕ is the angle between $\vec{F}$ and $\vec{s}$ (tail to tail). **Execute:** $W = Fs \cos \phi = (30)(5.0) \cos 40^\circ = 115 \text{ J}$. **Evaluate:** positive and below 150 J , as predicted; $\text{Nm} = \text{J}$. ✓

Perpendicular force: $\phi = 90^\circ$ gives $W = 0$ (carrying a bag at constant height).

But what if... what is the angle between the two vectors, when they are given by components rather than a picture? → **angle between vectors**.

Angle between two vectors



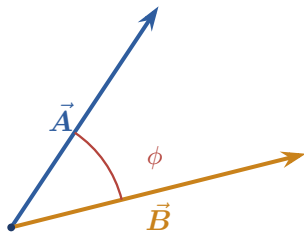
Lecturer example

$\vec{A} = 2\hat{i} + 3\hat{j}$ and $\vec{B} = 4\hat{i} + \hat{j}$. Find the angle ϕ between them.

Predict first

Both lie in quadrant I. Is ϕ acute or obtuse? Is $\vec{A} \cdot \vec{B}$ positive?

Angle between two vectors



Lecturer example

$\vec{A} = 2\hat{i} + 3\hat{j}$ and $\vec{B} = 4\hat{i} + \hat{j}$. Find the angle ϕ between them.

Predict first

Both lie in quadrant I. Is ϕ acute or obtuse? Is $\vec{A} \cdot \vec{B}$ positive?

Execute: $\vec{A} \cdot \vec{B} = (2)(4) + (3)(1) = 11$, $A = \sqrt{13} = 3.61$, $B = \sqrt{17} = 4.12$

$$\cos \phi = \frac{11}{(3.61)(4.12)} = 0.740 \Rightarrow \phi = 42.3^\circ$$

Evaluate: positive dot product, acute angle, as predicted. Check with directions:
 $\tan^{-1}(3/2) - \tan^{-1}(1/4) = 56.3^\circ - 14.0^\circ = 42.3^\circ$. ✓

One vector pair for the rest of the week

We will use these throughout Contact 3

$$\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}, \quad \vec{B} = -5\hat{i} + \hat{j} - \hat{k}$$

Your turn: find $\vec{A} \cdot \vec{B}$ and the angle ϕ between $\vec{A}$ and $\vec{B}$.

One vector pair for the rest of the week

We will use these throughout Contact 3

$$\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}, \quad \vec{B} = -5\hat{i} + \hat{j} - \hat{k}$$

Your turn: find $\vec{A} \cdot \vec{B}$ and the angle ϕ between $\vec{A}$ and $\vec{B}$.

$$\vec{A} \cdot \vec{B} = (1)(-5) + (9)(1) + (2)(-1) = 2$$

One vector pair for the rest of the week

We will use these throughout Contact 3

$$\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}, \quad \vec{B} = -5\hat{i} + \hat{j} - \hat{k}$$

Your turn: find $\vec{A} \cdot \vec{B}$ and the angle ϕ between $\vec{A}$ and $\vec{B}$.

$$\vec{A} \cdot \vec{B} = (1)(-5) + (9)(1) + (2)(-1) = 2$$

$$A = \sqrt{1 + 81 + 4} = \sqrt{86} = 9.27, \quad B = \sqrt{25 + 1 + 1} = \sqrt{27} = 5.20$$

$$\cos \phi = \frac{\vec{A} \cdot \vec{B}}{AB} = \frac{2}{(9.27)(5.20)} = 0.0415 \Rightarrow \phi = 87.6^\circ$$

Evaluate: small positive dot product $\Rightarrow$ angle just under 90° . ✓

But what if... a push perpendicular to a door's lever arm gives $\vec{r} \cdot \vec{F} = 0$, yet turns it most? $\rightarrow$ **cross product.**

Dot product and cross product compared

Dot product

$$\vec{A} \cdot \vec{B} = AB \cos \phi$$

- Result: a **scalar**
- Zero when the vectors are perpendicular
- Maximum magnitude when the vectors are parallel or antiparallel

Cross product

$$|\vec{A} \times \vec{B}| = AB \sin \phi$$

- Result: a **vector**
- Zero when the vectors are parallel or antiparallel
- Maximum magnitude when the vectors are perpendicular

The angle ϕ in both products

ϕ is the angle between the vectors when placed tail-to-tail, $0^\circ \leq \phi \leq 180^\circ$. **Do not use the internal angle of a head-to-tail vector triangle as ϕ** , such as the $\gamma = 120^\circ$ in the cyclist problem.

Question

How can two vectors produce a *third* vector?

The cross (vector) product: geometry first

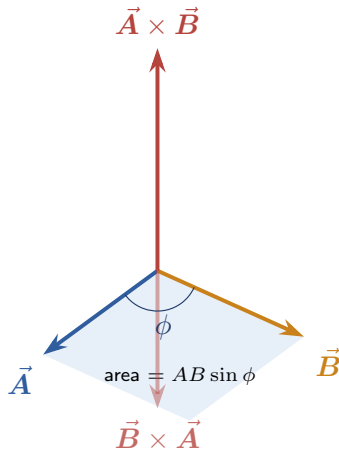
Question

How can two vectors produce a *third* vector?

Definition: cross product

- Magnitude: $|\vec{A} \times \vec{B}| = AB \sin \phi$ (area of the parallelogram)
- Direction: perpendicular to the plane of $\vec{A}$ and $\vec{B}$, given by the **right-hand rule**

The right-hand rule is on the next slide.



Using the right-hand rule

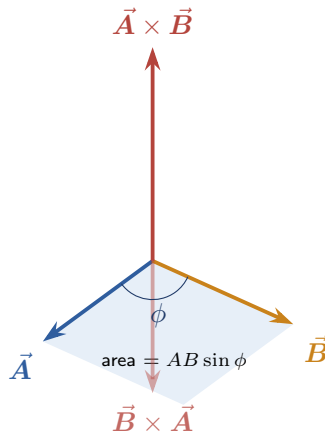
Right-hand rule

For $\vec{A} \times \vec{B}$:

1. Place $\vec{A}$ and $\vec{B}$ tail to tail.
2. Point the fingers of the right hand along $\vec{A}$.
3. Curl the fingers toward $\vec{B}$ through the smaller angle.
4. The thumb gives the direction of $\vec{A} \times \vec{B}$.

Caution

The order matters: $\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$. Swapping the vectors reverses the direction.



But what if... we swap the two vectors? Does the thumb stay? $\rightarrow$ **properties.**

Properties of the cross product

Predict first

What is $\vec{A} \times \vec{A}$? What happens if you swap the order?

Predict first

What is $\vec{A} \times \vec{A}$? What happens if you swap the order?

Rules: properties of the cross product

$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$ (the order matters)

Parallel or antiparallel vectors: $\sin \phi = 0 \Rightarrow \vec{A} \times \vec{B} = \vec{0}$

$\vec{A} \times \vec{B}$ is perpendicular to both $\vec{A}$ and $\vec{B}$

Rule: unit-vector cross products

Cyclic order $\hat{i} \rightarrow \hat{j} \rightarrow \hat{k} \rightarrow \hat{i}$ gives $+$: $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$.

Reversed order gives $-$: $\hat{j} \times \hat{i} = -\hat{k}$, $\hat{k} \times \hat{j} = -\hat{i}$, $\hat{i} \times \hat{k} = -\hat{j}$.

Also $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{0}$.

$\times$	$\hat{i}$	$\hat{j}$	$\hat{k}$
$\hat{i}$	$\vec{0}$	$\hat{k}$	$-\hat{j}$
$\hat{j}$	$-\hat{k}$	$\vec{0}$	$\hat{i}$
$\hat{k}$	$\hat{j}$	$-\hat{i}$	$\vec{0}$

row $\times$ column

EQUATION – cross product

Meaning. $\vec{A} \times \vec{B}$ is a vector perpendicular to both; its magnitude is largest when the vectors are perpendicular and zero when they are parallel.

Variables. A , B are the magnitudes; ϕ is the angle between the vectors tail to tail, $0^\circ \leq \phi \leq 180^\circ$.

Units. the product of the units: $\text{m} \times \text{m} = \text{m}^2$; for position and force, N m .

Interpretation. the magnitude is the area of the parallelogram on the two vectors; the direction is given by the right-hand rule; the order matters.

Example. the lecturer example on 30° vectors, then the pair $\vec{A}$, $\vec{B}$ used for the dot product.

Magnitude and direction

$$|\vec{A} \times \vec{B}| = AB \sin \phi$$

Direction: right-hand rule.

Predict first

When is $|\vec{A} \times \vec{B}|$ largest?

When is it zero?

Cross product from components

Rule: cross product from components

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y) \hat{i} + (A_z B_x - A_x B_z) \hat{j} + (A_x B_y - A_y B_x) \hat{k}$$

Memory aid: the determinant

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

The $\hat{j}$ term takes a minus sign when the determinant is expanded.

Connect: torque, Lecture 8

$$\vec{\tau} = \vec{r} \times \vec{F}$$

A force $\vec{F}$ applied at position $\vec{r}$ from a pivot. Torque is studied fully in Lecture 8.

But what if... we want a physical application? A force applied at a distance from a pivot. $\rightarrow$ torque.

TOOLKIT 7 – Cross product

WHEN TO USE: you need a vector perpendicular to two vectors, such as torque.

PROCEDURE: Magnitude: $|\vec{A} \times \vec{B}| = AB \sin \phi$.

Direction: right-hand rule; $\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$.

Unit vectors: cyclic $\hat{i} \rightarrow \hat{j} \rightarrow \hat{k} \rightarrow \hat{i}$ gives +;

reversed gives -; $\hat{i} \times \hat{i} = \vec{0}$.

Preview: torque, $\vec{\tau} = \vec{r} \times \vec{F}$.

CHECK: $(\vec{A} \times \vec{B}) \cdot \vec{A} = 0$ and $(\vec{A} \times \vec{B}) \cdot \vec{B} = 0$; parallel $\rightarrow \vec{0}$ ($\sin 0^\circ = \sin 180^\circ = 0$); perpendicular $\rightarrow$ maximum magnitude.

Components:

$$(A_y B_z - A_z B_y) \hat{i}$$

$$+(A_z B_x - A_x B_z) \hat{j}$$

$$+(A_x B_y - A_y B_x) \hat{k}$$

Question

$\vec{A}$ has magnitude 6 units along $+x$. $\vec{B}$ has magnitude 4 units in the xy -plane at 30° to $+x$. Find $\vec{A} \times \vec{B}$.

Predict first

Before calculating: which way will $\vec{A} \times \vec{B}$ point?

Question

$\vec{A}$ has magnitude 6 units along $+x$. $\vec{B}$ has magnitude 4 units in the xy -plane at 30° to $+x$. Find $\vec{A} \times \vec{B}$.

Predict first

Before calculating: which way will $\vec{A} \times \vec{B}$ point?

Magnitude:

$$AB \sin \phi = (6)(4) \sin 30^\circ = 12$$

Direction: right-hand rule from $+x$ towards $\vec{B}$ gives $+z$.

$$\vec{A} \times \vec{B} = 12 \hat{k}$$

Question

$\vec{A}$ has magnitude 6 units along $+x$. $\vec{B}$ has magnitude 4 units in the xy -plane at 30° to $+x$. Find $\vec{A} \times \vec{B}$.

Predict first

Before calculating: which way will $\vec{A} \times \vec{B}$ point?

Magnitude:

$$AB \sin \phi = (6)(4) \sin 30^\circ = 12$$

Direction: right-hand rule from $+x$ towards $\vec{B}$ gives $+z$.

$$\vec{A} \times \vec{B} = 12 \hat{k}$$

Check (components): $\vec{A} = 6\hat{i}$,
 $\vec{B} = 3.46\hat{i} + 2\hat{j}$
 $\Rightarrow (6)(2) - (0)(3.46) = 12$ along $\hat{k}$. ✓

Your turn: the same vector pair

Your turn

$\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}$, $\vec{B} = -5\hat{i} + \hat{j} - \hat{k}$. Find $\vec{A} \times \vec{B}$, then check it (Toolkit 7).

Your turn

$\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}$, $\vec{B} = -5\hat{i} + \hat{j} - \hat{k}$. Find $\vec{A} \times \vec{B}$, then check it (Toolkit 7).

$$(\vec{A} \times \vec{B})_x = A_y B_z - A_z B_y = (9)(-1) - (2)(1) = -11$$

$$(\vec{A} \times \vec{B})_y = A_z B_x - A_x B_z = (2)(-5) - (1)(-1) = -9$$

$$(\vec{A} \times \vec{B})_z = A_x B_y - A_y B_x = (1)(1) - (9)(-5) = 46$$

$$\vec{A} \times \vec{B} = -11\hat{i} - 9\hat{j} + 46\hat{k}$$

Your turn

$\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}$, $\vec{B} = -5\hat{i} + \hat{j} - \hat{k}$. Find $\vec{A} \times \vec{B}$, then check it (Toolkit 7).

$$(\vec{A} \times \vec{B})_x = A_y B_z - A_z B_y = (9)(-1) - (2)(1) = -11$$

$$(\vec{A} \times \vec{B})_y = A_z B_x - A_x B_z = (2)(-5) - (1)(-1) = -9$$

$$(\vec{A} \times \vec{B})_z = A_x B_y - A_y B_x = (1)(1) - (9)(-5) = 46$$

$$\vec{A} \times \vec{B} = -11\hat{i} - 9\hat{j} + 46\hat{k}$$

Perpendicularity check (Toolkit 7):

$$(\vec{A} \times \vec{B}) \cdot \vec{A} = -11 - 81 + 92 = 0 \quad \checkmark \qquad (\vec{A} \times \vec{B}) \cdot \vec{B} = 55 - 9 - 46 = 0 \quad \checkmark$$

Decide before calculating

1. Is $\vec{A} \cdot \vec{B}$ a scalar or a vector? Can it be negative?
2. Can $\vec{A} \cdot \vec{B} = 0$ for two non-zero vectors? Can $\vec{A} \times \vec{B} = \vec{0}$?
3. Is $\vec{A} \times \vec{B}$ ever parallel to $\vec{A}$?

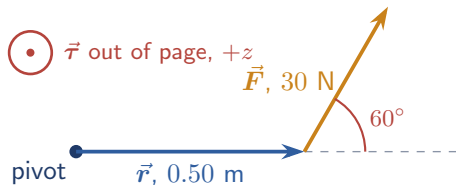
Decide before calculating

1. Is $\vec{A} \cdot \vec{B}$ a scalar or a vector? Can it be negative?
2. Can $\vec{A} \cdot \vec{B} = 0$ for two non-zero vectors? Can $\vec{A} \times \vec{B} = \vec{0}$?
3. Is $\vec{A} \times \vec{B}$ ever parallel to $\vec{A}$?

Answers

1. **Scalar.** Yes, when $\phi > 90^\circ$.
2. **Yes to both.** Perpendicular vectors give $\vec{A} \cdot \vec{B} = 0$; parallel or antiparallel vectors give $\vec{A} \times \vec{B} = \vec{0}$.
3. **Never.** $\vec{A} \times \vec{B}$ is perpendicular to both $\vec{A}$ and $\vec{B}$ (check: its dot product with each is zero).

Torque: the cross product in action



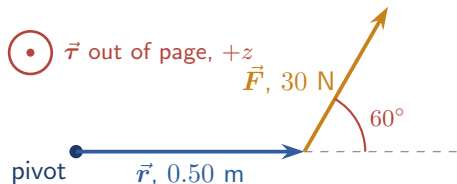
Lecturer example

A wrench is pivoted at one end. A force $F = 30 \text{ N}$ acts at $r = 0.50 \text{ m}$ from the pivot, at 60° to $\vec{r}$ (in the page). Find $\vec{\tau} = \vec{r} \times \vec{F}$.

Predict first

Is τ larger or smaller than $rF = 15 \text{ N m}$?
Which way does $\vec{\tau}$ point?

Torque: the cross product in action



Lecturer example

A wrench is pivoted at one end. A force $F = 30 \text{ N}$ acts at $r = 0.50 \text{ m}$ from the pivot, at 60° to $\vec{r}$ (in the page). Find $\vec{\tau} = \vec{r} \times \vec{F}$.

Predict first

Is τ larger or smaller than $rF = 15 \text{ N m}$?
Which way does $\vec{\tau}$ point?

Execute: $\tau = rF \sin \phi = (0.50)(30) \sin 60^\circ = 13.0 \text{ N m}$. Right hand: fingers along $\vec{r}$, curled toward $\vec{F}$, thumb out of the page: $+z$.

Evaluate: $13.0 < 15$, as predicted; $\tau = 0$ if $\vec{F} \parallel \vec{r}$ and $\tau = rF$ if $\vec{F} \perp \vec{r}$. ✓

But what if... a problem uses several of these ideas at once: adding, resolving, dot and cross products? Which tool, and when? → **integrated applications**.

Integrated problem: pulling a crate

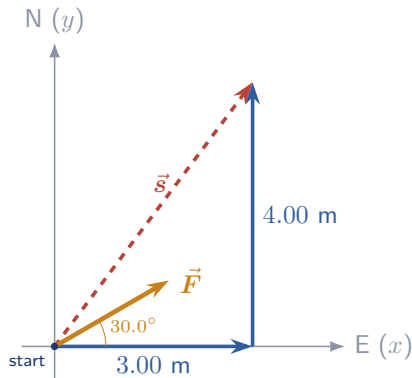
Application problem

A worker pulls a crate across a level floor with a rope. The rope force is 50.0 N at 30.0° north of east. The crate moves 3.00 m east, then 4.00 m north.

- (a) Find the crate's displacement $\vec{s}$.
- (b) Find the work done by the rope force, $W = \vec{F} \cdot \vec{s}$.
- (c) The same force acts at $\vec{r} = (0.800\hat{i} + 0.600\hat{j})\text{ m}$ from a pivot. Find the torque $\vec{\tau} = \vec{r} \times \vec{F}$.

Predict first

Sketch first. Is W positive, zero or negative?
Does $\vec{\tau}$ point along $+z$ or $-z$?



top view; $F = 50.0\text{ N}$; z points out of the page

Identify: targets $\vec{s}$, W and $\vec{\tau}$. **Set up:** x east, y north, z up; $\vec{F}$ at $\theta = 30.0^\circ$ (Toolkit 1). Predict: $\phi < 90^\circ$, so $W > 0$.

Identify: targets $\vec{s}$, W and $\vec{\tau}$. **Set up:** x east, y north, z up; $\vec{F}$ at $\theta = 30.0^\circ$ (Toolkit 1). Predict: $\phi < 90^\circ$, so $W > 0$.

Execute.

(a) Add the legs (Toolkit 5):

$$\vec{s} = (3.00\hat{i} + 4.00\hat{j}) \text{ m}$$

$$s = 5.00 \text{ m at } 53.1^\circ \text{ north of east}$$

Resolve the force (Toolkit 3):

$$F_x = 50.0 \cos 30.0^\circ = 43.3 \text{ N}$$

$$F_y = 50.0 \sin 30.0^\circ = 25.0 \text{ N}$$

(b) Dot product (Toolkit 6):

$$W = (43.3)(3.00) + (25.0)(4.00) = 230 \text{ J}$$

(c) Cross product (Toolkit 7); only the $\hat{k}$ term survives:

$$\tau_z = r_x F_y - r_y F_x = (0.800)(25.0) - (0.600)(43.3)$$

$$\vec{\tau} = -5.98 \hat{k} \text{ N m}$$

Identify: targets $\vec{s}$, W and $\vec{\tau}$. **Set up:** x east, y north, z up; $\vec{F}$ at $\theta = 30.0^\circ$ (Toolkit 1). Predict: $\phi < 90^\circ$, so $W > 0$.

Execute.

(a) Add the legs (Toolkit 5):

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$$\tau_z = r_x F_y - r_y F_x = (0.800)(25.0) - (0.600)(43.3)$$

$$\vec{\tau} = -5.98 \hat{k} \text{ N m}$$

Evaluate (verification):

$$Fs \cos \phi = (50.0)(5.00) \cos 23.1^\circ = 230 \text{ J } \checkmark$$

$$rF \sin \phi = (1.00)(50.0) \sin 6.9^\circ = 5.98 \text{ N m } \checkmark$$

Right-hand rule from $\vec{r}$ (36.9°) to $\vec{F}$ (30.0°):

clockwise seen from above, so $-z$ $\checkmark$

$W > 0$, as predicted $\checkmark$

Your turn: an integrated problem

Independent application

A rope force of 40.0 N acts at 60.0° north of east. The object moves 2.00 m north, then 2.00 m west.

(a) Find $\vec{s}$. (b) Find the work done by the force. (c) The force acts at $\vec{r} = 0.500\hat{i}$ m from a pivot; find $\vec{\tau}$.

Verify (b) with $Fs \cos \phi$ and (c) with $rF \sin \phi$ and the right-hand rule.

Predict first

Sketch first. Which quadrant is $\vec{s}$ in? Is ϕ less than 90° ?

Your turn: an integrated problem

Independent application

A rope force of 40.0 N acts at 60.0° north of east. The object moves 2.00 m north, then 2.00 m west.

(a) Find $\vec{s}$. (b) Find the work done by the force. (c) The force acts at $\vec{r} = 0.500\hat{i}$ m from a pivot; find $\vec{\tau}$.

Verify (b) with $Fs \cos \phi$ and (c) with $rF \sin \phi$ and the right-hand rule.

Predict first

Sketch first. Which quadrant is $\vec{s}$ in? Is ϕ less than 90° ?

(a) $\vec{s} = (-2.00\hat{i} + 2.00\hat{j})$ m, $s = 2.83$ m at 45° north of west

(b) $F_x = 20.0$ N, $F_y = 34.6$ N:

$$W = (20.0)(-2.00) + (34.6)(2.00) = 29.3 \text{ J}$$

Your turn: an integrated problem

Independent application

A rope force of 40.0 N acts at 60.0° north of east. The object moves 2.00 m north, then 2.00 m west.

(a) Find $\vec{s}$. (b) Find the work done by the force. (c) The force acts at $\vec{r} = 0.500\hat{i}$ m from a pivot; find $\vec{\tau}$.

Verify (b) with $Fs \cos \phi$ and (c) with $rF \sin \phi$ and the right-hand rule.

Predict first

Sketch first. Which quadrant is $\vec{s}$ in? Is ϕ less than 90° ?

(a) $\vec{s} = (-2.00\hat{i} + 2.00\hat{j})$ m, $s = 2.83$ m at 45° north of west

(b) $F_x = 20.0$ N, $F_y = 34.6$ N:

$$W = (20.0)(-2.00) + (34.6)(2.00) = 29.3 \text{ J}$$

Check: $\phi = 135^\circ - 60^\circ = 75^\circ$;

$$(40.0)(2.83) \cos 75^\circ = 29.3 \text{ J} \checkmark$$

(c) $\tau_z = r_x F_y - r_y F_x = (0.500)(34.6) - 0 = 17.3$, so $\vec{\tau} = 17.3 \hat{k}$ N m

Check: $(0.500)(40.0) \sin 60^\circ = 17.3$ N m;
counter-clockwise from above $\checkmark$

Common mistakes and the correct habit

Mistake

Using the internal triangle angle in $\vec{A} \cdot \vec{B}$ or $\vec{A} \times \vec{B}$.

Writing a dot product as a vector, or a cross product as a number.

Forgetting that $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$.

Swapping cos and sin between the two products.

A wrong sign on the $\hat{j}$ term of the cross product.

Taking $\hat{i} \cdot \hat{j} = 1$.

Instead

ϕ is the angle tail to tail, $0^\circ \leq \phi \leq 180^\circ$.

Dot gives a scalar; cross gives a vector.

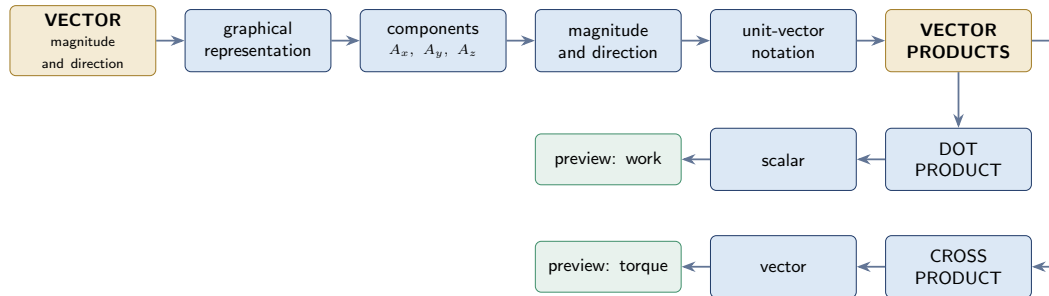
Use the right-hand rule and keep the order.

Test with the special cases: perpendicular and parallel.

Use the determinant, then check $(\vec{A} \times \vec{B}) \cdot \vec{A} = 0$.

Like unit vectors give 1, unlike give 0.

Week 1 concept map



Key ideas

- **A vector is a mathematical object used to represent a physical quantity; its components depend on the axes chosen.**
- A_x is a number; $A_x \hat{i}$ is the component vector.
- Magnitudes do not simply add: $|A - B| \leq |\vec{A} + \vec{B}| \leq A + B$.
- **Graphs give geometric understanding; components provide a systematic way to calculate accurately.**
- Magnitude/direction and components describe the same vector.
- **The dot product measures the component of one vector along another; the cross product produces a vector perpendicular to both input vectors.** Their magnitudes are $AB \cos \phi$ and $AB \sin \phi$, respectively.
- Habit: sketch, predict, calculate, evaluate.

Reference tools: where to look

Equations, procedures: the next slides (core equations, master toolkit, seven toolkits).

Definitions: the vocabulary and equation-anatomy slides.

Common mistakes: the last slide of each contact.

What you should remember: the chain, Contact 1 (part 1)

Concept	Limitation that motivates the next	Leads to
Problem-solving framework	needs quantities to work with: “5 m away” does not find a bag	Magnitude
Magnitude	every point on a 5 m circle fits	Direction
Direction	mass has none; some quantities need both	Scalars, vectors
Scalars, vectors	400 m round a track: distance cannot say where you are	Displacement
Displacement	how to name, draw and compare it?	Vocabulary
Vocabulary	two displacements in a row: what is the net?	Opening problem
Opening problem	1 km + 2 km as numbers ignores direction	Graphical addition
Graphical addition	how big is $\vec{R}$? Try one line first	Parallel, antiparallel

What you should remember: the chain, Contact 1 (part 2)

Concept	Limitation that motivates the next	Leads to
Parallel, antiparallel	off the line, $3\text{ m} + 4\text{ m} \neq 7\text{ m}$	Magnitudes do not add
Magnitudes do not add	three or more vectors?	Associativity
Associativity	undo an addition?	Subtraction
Subtraction	drawings are only as accurate as a ruler	90° vectors
90° vectors	why does the hypotenuse follow a rule?	Pythagoras
Pythagoras	solves the opening problem; at 60° it gives 3.61 km, yet at least 4.00 km is needed	Limitation, general triangles
General triangles	four legs mean three triangles	Components

What you should remember: the chain, Contact 2

Concept	Limitation that motivates the next	Leads to
Why components	how much of $\vec{A}$ lies along each axis?	Components
Components	given A and θ , how to get the parts?	Resolving
Signs, quadrants	angles measured from other axes	Resolving carefully
Resolving	backwards: $\tan^{-1}$ gives the same value for opposite vectors	Reconstructing
Reconstructing	many vectors, each with components	Adding vectors
Adding vectors	displacements given as magnitudes and compass directions	Multiple vectors
Multiple vectors	writing components in words is long	Unit vectors
Unit vectors	only planar examples so far	3D vectors
3D vectors	we can add and subtract, but not multiply	Dot product

What you should remember: the chain, Contact 3

Concept	Limitation that motivates the next	Leads to
Dot product	which angle is ϕ ? Tail to tail, not the triangle angle γ	The angle ϕ
The angle ϕ	what does a dot product do for a force and a displacement?	Work
Work	the angle between vectors given by components	Angle between vectors
Angle between vectors	a perpendicular push turns a door most, but $\vec{r} \cdot \vec{F} = 0$	Cross product
Cross product	does the order matter?	Right-hand rule, properties
Cross product	a force applied at a distance from a pivot	Torque
Torque	problems that combine several tools	Integrated applications

Reference tools: core equations

Idea	Equation	Conditions and units
Along a line	$R = A + B \quad \text{or} \quad R = A - B $	one line only
Perpendicular	$R = \sqrt{A^2 + B^2}, \quad \tan \theta = B/A$	90° between the vectors
Cosine rule	$R^2 = A^2 + B^2 - 2AB \cos \gamma$	γ : internal angle opposite R
Components	$A_x = A \cos \theta, \quad A_y = A \sin \theta$	θ from $+x$; units of $\vec{A}$
Magnitude, direction	$A = \sqrt{A_x^2 + A_y^2}, \quad \theta_{\text{ref}} = \tan^{-1} A_y/A_x $	place by quadrant
Addition	$R_x = \sum A_x, \quad R_y = \sum A_y$	same units for all terms
Unit-vector form	$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$	$\hat{i}, \hat{j}, \hat{k}$ dimensionless
Dot product	$AB \cos \phi = A_x B_x + A_y B_y + A_z B_z$	scalar; product of units
Cross product	$ \vec{A} \times \vec{B} = AB \sin \phi$	vector; right-hand rule
Cross, components	$(A_y B_z - A_z B_y) \hat{i} + (A_z B_x - A_x B_z) \hat{j} + (A_x B_y - A_y B_x) \hat{k}$	$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$

MASTER TOOLKIT – For a general vector problem

1. **IDENTIFY** What is being asked?
2. **SKETCH** Draw the physical situation.
3. **CHOOSE AXES** Select useful perpendicular coordinates.
4. **PREDICT** Estimate signs, quadrant, size and direction.
5. **RESOLVE** Convert vectors into components if necessary.
6. **COMBINE** Perform the required vector operation.
7. **RECONSTRUCT** Find magnitude and direction if required.
8. **EVALUATE** Check units, signs, quadrant, magnitude, direction and physical reasonableness.

If the result disagrees with the sketch or prediction, investigate before accepting the answer.

Inside the course framework

IDENTIFY (step 1): what is being asked?

SET UP (steps 2–4): sketch, choose axes, predict.

EXECUTE (steps 5–7): resolve, combine, reconstruct.

EVALUATE (step 8): check the result.

THE SEVEN TOOLKITS

- | | | |
|---|--------------------------------|--|
| 1 | Directions and angles | directional language, standard angle, quadrant, reference angle, compass directions |
| 2 | Vector triangles | head to tail, internal angle, Pythagoras or sine and cosine rules, $ A - B \leq R \leq A + B$ |
| 3 | Components | choose axes $\rightarrow$ sketch $\rightarrow$ predict signs $\rightarrow$ resolve |
| 4 | Reconstructing a vector | magnitude; direction by quadrant, reference angle, placement, check |
| 5 | Adding by components | RESOLVE $\rightarrow$ COMBINE $\rightarrow$ RECONSTRUCT |
| 6 | Dot product | two forms, sign check, angle between vectors; preview: work |
| 7 | Cross product | magnitude, right-hand rule, unit vectors, components, perpendicularity check; preview: torque |

Extension: direction angles in 3D are in the supplementary notes.

Checklist: can I ...

- ☐ tell a scalar from a vector?
- ☐ add and subtract vectors graphically (parallel, antiparallel, perpendicular)?
- ☐ use the sine and cosine rules on a vector triangle?
- ☐ resolve a vector with the correct signs?
- ☐ determine the direction of a vector, including its correct quadrant?
- ☐ add several vectors by components?
- ☐ write a vector with $\hat{i}, \hat{j}, \hat{k}$?
- ☐ use IDENTIFY → SET UP → EXECUTE → EVALUATE?
- ☐ calculate a dot product?
- ☐ determine whether a dot product is positive, zero or negative?
- ☐ use the dot product to find the angle between two vectors?
- ☐ calculate a cross product and determine its direction?
- ☐ use the right-hand rule?
- ☐ explain when a cross product is zero?

Next

Problem sheet 1 on Sakai: extra component, resultant, dot and cross problems, plus challenge problems. Supplementary notes cover linear independence and direction angles in 3D (extension).